

Talia Maeve O'Shea

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Education

University of Wisconsin-Madison, Ph.D. candidate in Astronomy, in progress	Aug 2023 – Present
• NSF Graduate Research Fellow	
University of Wisconsin-Madison, M.S. in Astronomy	Aug 2023 – May 2025
Wellesley College, B.A. in Astrophysics (Honors) and History, <i>Summa Cum Laude</i>	Sept 2019 – May 2023

Publications

O'Shea T.M., et al., *Predicting the ram pressure bent-tail radio galaxy population inferred from cosmological simulations*. In prep, expected submission July 2026.

O'Shea T.M., Heinz S., Soares-Furtado M., Igo Z., Merloni A., *Shooting for the stars: Jet-mode feedback and AGN jet deceleration from stellar mass loading*. ApJ, **995** 106. [arXiv:2510.26881](https://arxiv.org/abs/2510.26881).

O'Shea T.M., Borrow J., O'Neil S., Vogelsberger M., *Dynamical friction and measurements of the splashback radius in galaxy clusters*. OJAp, **8**. [arXiv:2405.18468](https://arxiv.org/abs/2405.18468).

Research Positions

Visiting Graduate Researcher, Dr. Volker Springel; Max Planck Institute for Astrophysics, Garching DE.	Apr 2026 - May 2026
Graduate Researcher, Dr. Sebastian Heinz; University of Wisconsin-Madison, Madison WI.	Aug 2023 - Present
Undergraduate Researcher, Dr. Mark Vogelsberger and Dr. Josh Borrow; MIT, Cambridge MA.	May 2022 - Aug 2023
Undergraduate Researcher, Dr. James Battat and Dr. Chris Arumainayagam; Wellesley College, Wellesley MA.	Jan 2021 - Oct 2022

Presentations

Invited Talks

1. **Max Planck Institute for Astronomy (MPIA)** Galaxy Coffee, "Differentiating direct and preventative AGN jet feedback via stellar wind mass-loading," Heidelberg DE, May 2026.
2. **Max Planck Institute for Astrophysics (MPA)** Cosmology and Galaxies Seminar, "Differentiating direct and preventative AGN jet feedback via jet deceleration," Garching DE, April 2026.
3. **Indiana University**, "Shooting for the stars: Decelerating AGN jets by entraining stellar winds," February 2026.
4. **University of Massachusetts-Amherst**, "Decelerating AGN jets with stellar winds," August 2025.
5. **University of Maryland**, "Dynamical friction and measurements of galaxy cluster size," November 2024.
6. **University of Pennsylvania**, "Dynamical friction and measurements of galaxy cluster size," October 2023.
7. **Wellesley College Tanner Conference**, *Co-Presenter*, "On Becoming a Scientist: Finding Our Passion Through Wellesley's Summer Research Program," Fall 2021.

Contributed Talks

1. **Texas Symposium on Relativistic Astrophysics**, "Shooting for the stars: Jet-mode feedback and AGN jet deceleration from stellar mass-loading," December 2025.
2. **Midwest Magnetic Fields Workshop**, "Shooting for the stars: Jet-mode feedback and AGN jet deceleration from stellar mass-loading," July 2025.
3. **Wellesley College Ruhlman Conference**, "Dynamical friction in galaxy clusters," April 2023.

4. **American Astronomical Society 240th Conference**, “Low-energy (<20 eV) electrons in Geant4 Monte Carlo Simulations,” Summer 2022.
5. **American Astronomical Society 238th Conference**, “Investigating the relative importance of low-energy (<20 eV) electrons in astrochemistry via Monte Carlo simulations,” Summer 2021 (virtual).

Posters

1. *Upcoming*: **36th Annual Wisconsin Space Conference**, “Black hole jets and their environments,” August 2026.
2. **American Astronomical Society 244th Conference**, “The impact of main sequence stellar mass loading on AGN jet feedback,” June 2024.
3. **Beckman Symposium**, “Dynamical friction has limited impact on galaxy cluster splashback radii,” August 2023.
4. **Wellesley College Science Center Summer Research Program**, “Dynamical friction has limited impact on measurements of galaxy cluster size,” Summer 2022.
5. **Keck Northeast Astronomy Consortium**, “Assessing the influence of dynamical friction on galaxy cluster splashback radius,” Fall 2022.
6. **Wellesley College Science Center Summer Research Program**, “The role of low-energy (<20 eV) electrons in radiation chemistry,” Summer 2021, *Co-Presenter*.

Outreach Talks

1. **Universe in the Park**, Blue Mounds WI; June 2026.
2. **Madison WI Astronomy on Tap**, “Nancy Drew and the Secrets of Supermassive Black Holes,” November 2025.
3. **Universe in the Park**, Sheboygan WI; August 2025.
4. **Universe in the Park**, St. Croix Falls, Chippewa Falls WI; July 2025.
5. **Universe in the Park**, Superior WI; August 2024.

Mentoring Experience

Direct mentor to:

- Rachel G., UW-Madison undergraduate, Livingston Scholar (Jan 2025 - Present)
- Jack C., UW-Madison undergraduate (Apr 2026 - Present)

Teaching Experience

Teaching Assistant , ASTRON 103 (Introductory Astronomy); UW-Madison	Sep 2023 – Dec 2023
Learning Assistant , PHYS 106 (Introductory Electromagnetism); Wellesley College	Sept 2021 – May 2023

Professional Development

Developer of Research Mentorship Community ; Madison, WI	Spring 2026
Designing and leading a pilot research mentorship program for early career researchers in Astronomy and Physics.	
Code/Astro Workshop ; Evanston, IL	Aug 2025
Delta Program Research Mentor Training ; Madison, WI	May 2024

Grants and Awards

Funding

National Science Foundation Graduate Research Fellow (\$150,000)	Jun 2025 - Present
Blair Savage Travel Fund (\$750)	Jan 2026
Wisconsin Space Grant Consortium Graduate Research Fellowship (\$6,000)	Sep 2025 - May 2026

Beckman Scholar , Arnold and Mabel Beckman Foundation (\$21,000)	2021 - 2023
Massachusetts Space Grant (\$5,000)	2022
National Merit Scholarship , Textron Corporation (\$5,000)	2019 - 2023

Computing Time

National Science Foundation ACCESS “EXPLORE” Allocation PHY250213	Jul 2025 - Jul 2026
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Awards

Stebbins Award , UW-Madison Astronomy Department	April 2026
Phyllis J. Fleming Prize , Wellesley College Physics Department	May 2023
Whiting Medal , Wellesley College Astronomy Department	November 2023

Service

Department

Graduate Student Chair ; UW-Madison Astronomy Department	Jun 2026 - Present
Curriculum Committee ; UW-Madison Astronomy Department	Sep 2025 - May 2026
Undergraduate Symposium Planning Committee ; UW-Madison Astronomy Department	Apr 2025 - May 2025
Peer Mentor Chair ; UW-Madison Astronomy Department Running peer mentorship program for first-year graduate students.	Aug 2024 – Jun 2026
Inclusive Excellence Committee Planning Task Force ; UW-Madison Astronomy Department	Feb 2024 – June 2024
Pinanski Prize Committee Student Representative ; Wellesley College	Apr 2023
Society of Physics Students ; Wellesley College Served variously as Co-President, Secretary, Department Liaison, and first-year representative, to organize department events and build community. Represented student voices at department meetings.	Dec 2019 - May 2023

Other

Reviewer , Student Spaceflight Experiments Program	Nov 2025
Reviewer , UW-Madison REU program	Feb 2026

Professional memberships

American Astronomical Society	Oct 2021 - Present
Phi Beta Kappa	May 2023 - Present
Sigma Xi Nominee	May 2023

Skills

Technical: Python, C, Unix, LaTeX, Arepo, Geant4
Languages: English (native), Mandarin (fluent; Massachusetts Seal of Biliteracy)
Citizenship: USA, Ireland